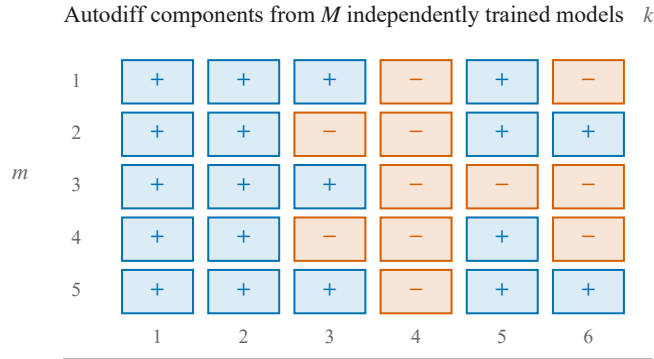


Reference-calibrated selective verification of surrogate Jacobians

Rank components after training; spend reference simulations only where calibrated evidence is insufficient.

a Cross-retrain evidence



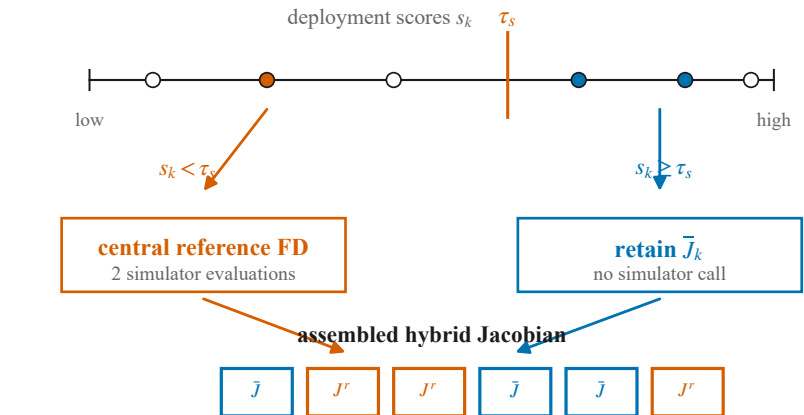
Candidate component scores

mean-aligned sign a_k
signal-to-noise $u_k = |\bar{J}_k|/\hat{\sigma}_k$
magnitude $m_k = |\bar{J}_k|$

b Calibrate, then deploy

Reference-labelled calibration queries

Choose score and τ_s for the intended loss; abstain if the risk screen fails.



c Measured operating point ($a_k \geq 0.9$)

benchmark	coverage	risk evals/query
TMM near-saturated	95.9%	1.5% 0.90/22
TMM stressed	74.0%	13.4% 5.72/22
Heat/Poisson stressed	79.9%	4.9% 6.44/32

Score choice depends on the loss

ROC AUC SNR: both TMM; magnitude: heat
fixed-budget error sign: lowest in all three
 $R_{FT} = P(\text{sign wrong} \mid \text{accepted})$
cost column: selective / complete central FD

Scope boundary

common positive rescaling of all members at fixed $k \Rightarrow$ unchanged sign agreement and SNR; m_k changes

shared bias $\Rightarrow$ high agreement can still be wrong

Evidence screen: 3,580/3,580 external labels passed step/grid checks; unresolved antenna derivatives were excluded.

calibrated triage $\neq$ correctness certificate